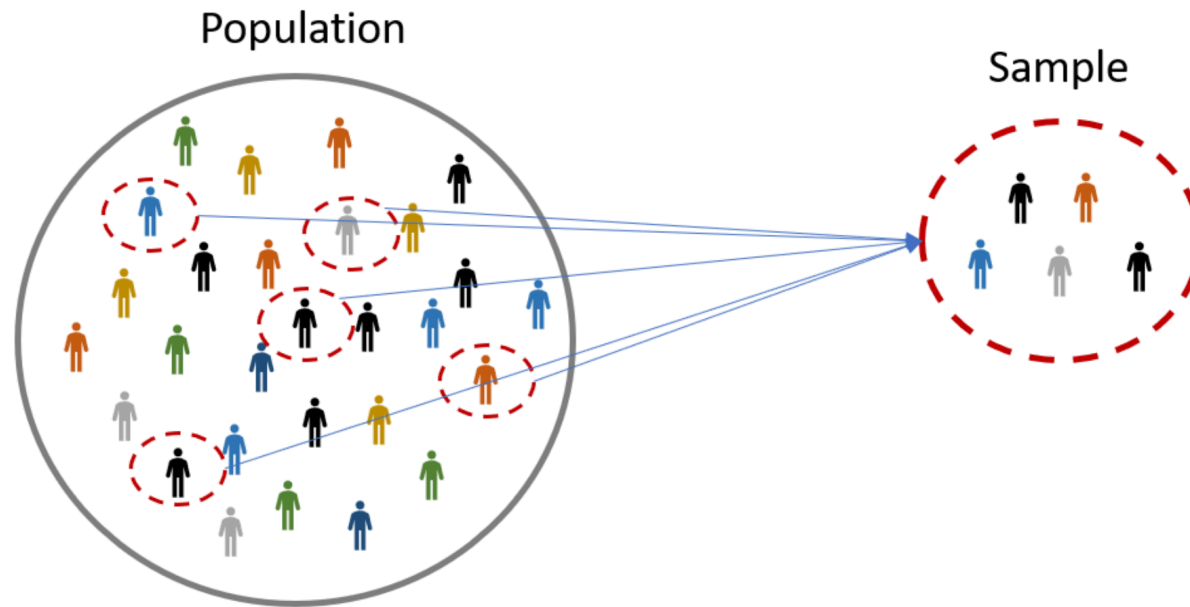


Simulation and Resampling (Solutions)

Using R to simulate a random process

In statistics, we study a **sample** to learn about a larger **population** (e.g. all American men).



This introduces *randomness*.

Probability (and stats) is the study of randomness.

A **random process** is any repeatable process whose outcome is uncertain.

- Rolling a die
- Flipping a coin
- Measuring a random person's height

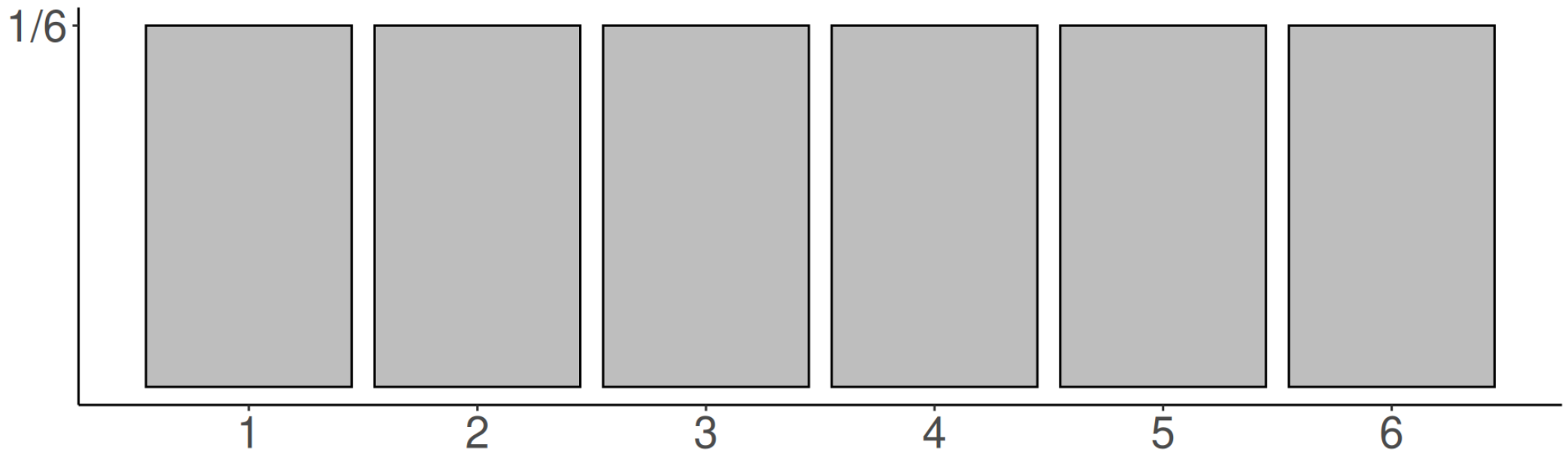
A population represents one type of random process.



How can we study a random process?

- **Modeling:** Make assumptions about its structure (classical statistics)
- **Analysis:** with pure math and probability, if the process is known
- **Simulation:** sample from the process a million bajillion times, see what patterns emerge

Some random processes are simple to study analytically *or* with simulation, e.g. a single dice roll.



This is the **distribution**. The average is 3.5, min is 1, etc.

Some processes are *possible* but *cumbersome* to study analytically, so we prefer simulation.

A Dungeons & Dragons character has six ability scores. Each one. e.g. Strength, is calculated as:

- Roll four six-sided dice. Sum up the highest three rolls.

What is the distribution of Strength scores?



Let's look at a more complex example. A D&D character has six scores, not just one:

average score: 12

STR	DEX	CON
16	11	14
INT	WIS	CHA
10	13	8



Each character has a different mean/average of their six scores.

average score: 11.17

STR	DEX	CON
9	10	13
INT	WIS	CHA
17	5	13



average score: 13

STR	DEX	CON
10	13	14

INT	WIS	CHA
10	15	16



Exercises

Re-create the simulated histogram, but with the random process “generate six scores and take the average”

- Complete the code for `average_scores` to get 10000 example characters and their average of the six scores.
- Plot the averages with `geom_histogram()`
- Compare this plot to the one of strength scores. What do you notice? What does the x axis look like?



Solutions/Hints

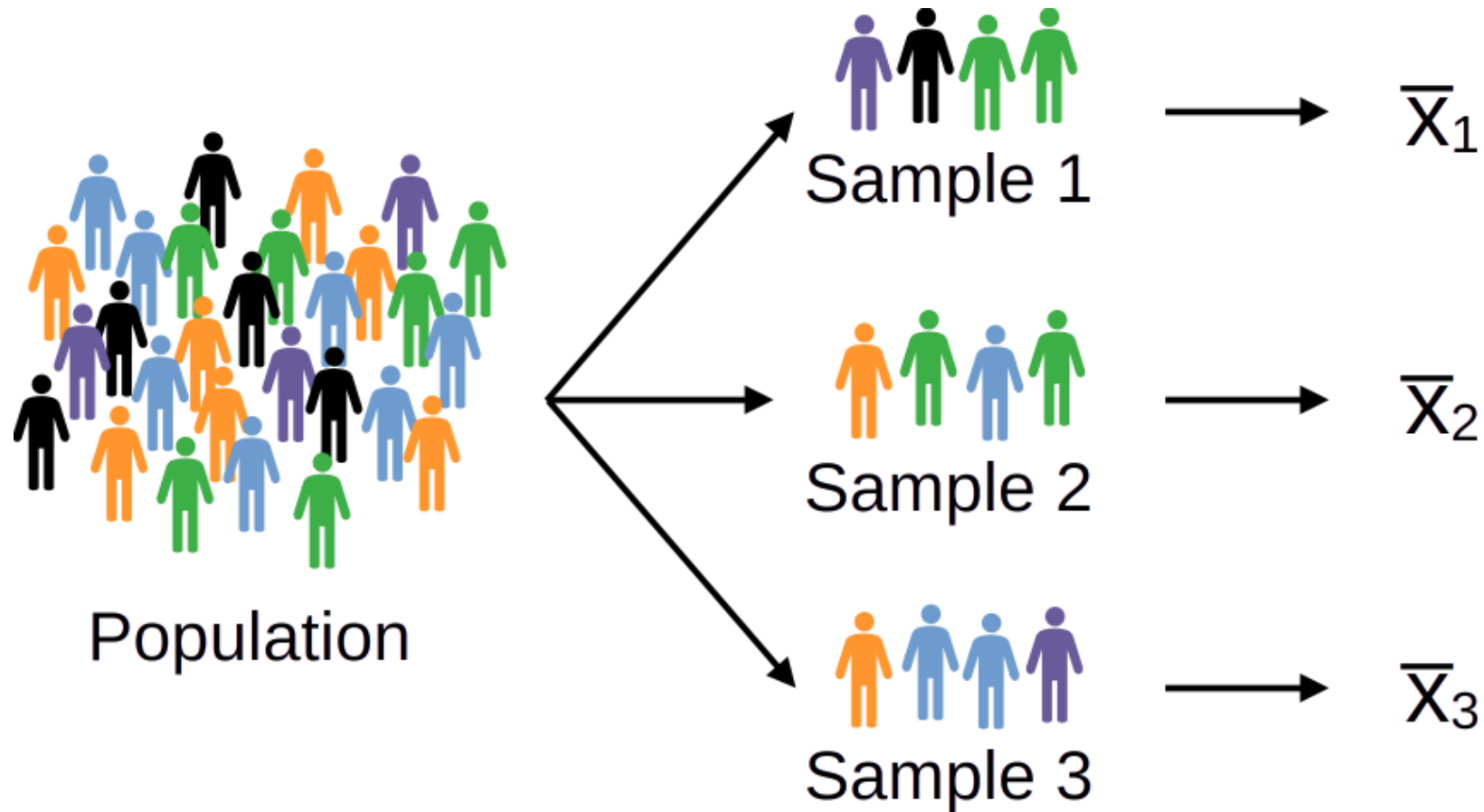
- `replicate(10000, mean(all_character_scores()))`
- Feel free to copy the histogram code. In this case, the x-axis refers to *averages*.
- Compared to just looking at raw strength score, this histogram is more concentrated and more symmetric.

This histogram is a **sampling distribution** of the mean.

1. Start with original process (e.g. D&D ability score)
2. Take a sample ($n = 6$)
3. Find a statistic (e.g. mean)
4. Repeat this process (10000 times)

Result: We get to study the behavior (sampling distribution) of the sample mean $\bar{x}$.

Each time we sample from the population/random process, we get a different sample mean.



Why do we care about sampling distributions?

Consider the penguin example. The mean flipper length of the Adelie penguins in our data is 190.1 mm.

Is this a good estimate of the true mean?

Do we have enough data to make a useful conclusion?

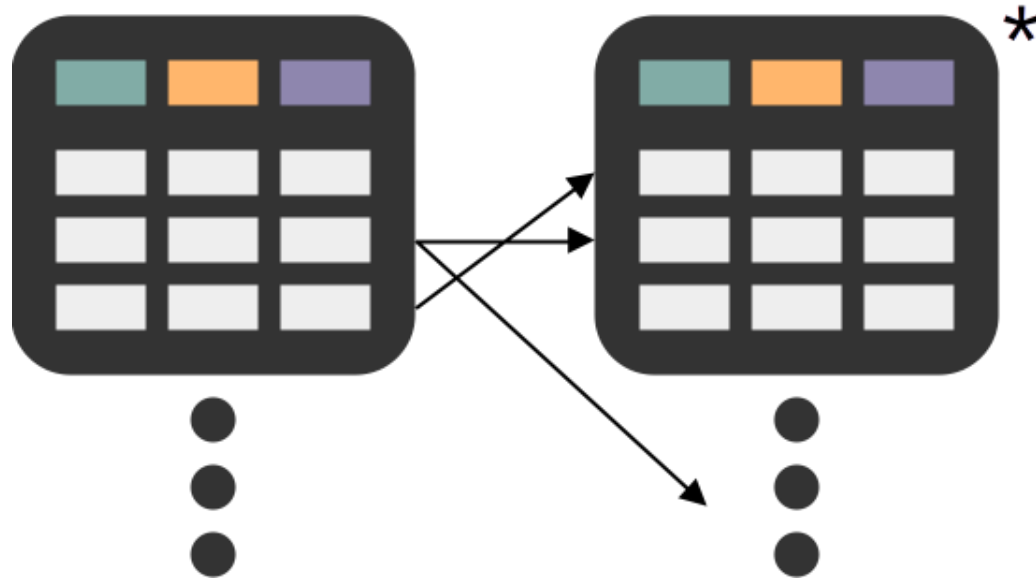
If we repeated the study, would the new sample mean be close to 190.1?

Consider the sampling distribution of mean flipper length.

1. Start with original process... Uh.....

- ~~Analysis: with pure math and probability~~
- **Simulation:** sample from the process a million bajillion times, see what patterns emerge - maybe????

We can't "see" the full population. How can we make simulation work?



We're going to resample from the *data* as an approximation to resampling from the *population*.

Let n be the original number of samples. Randomly sample n rows from the data, *with* replacement. This is our “new” data.

Exercises

Briefly discuss with your neighbor:

- Why do we want to have the same n from the original data for our resample?
- Why do we sample with replacement?
- Will every row from the original data appear in the resample? Can rows appear more than once?



Solutions/Hints

- `replicate(10000, mean(all_character_scores()))`
- Feel free to copy the histogram code. In this case, the x-axis refers to *averages*.
- Compared to just looking at raw strength score, this histogram is more concentrated and more symmetric.



Replication time!

Exercises

This isn't limited to the mean - we can study the sampling distribution of any summary statistic!

- Repeat the resampling exercise, but study the *median* flipper length for *Gentoo* penguins.
- Create a histogram of resampled medians.
- A different study suggests the median flipper length is 215 mm. Does your result agree with this?



Solutions/Hints

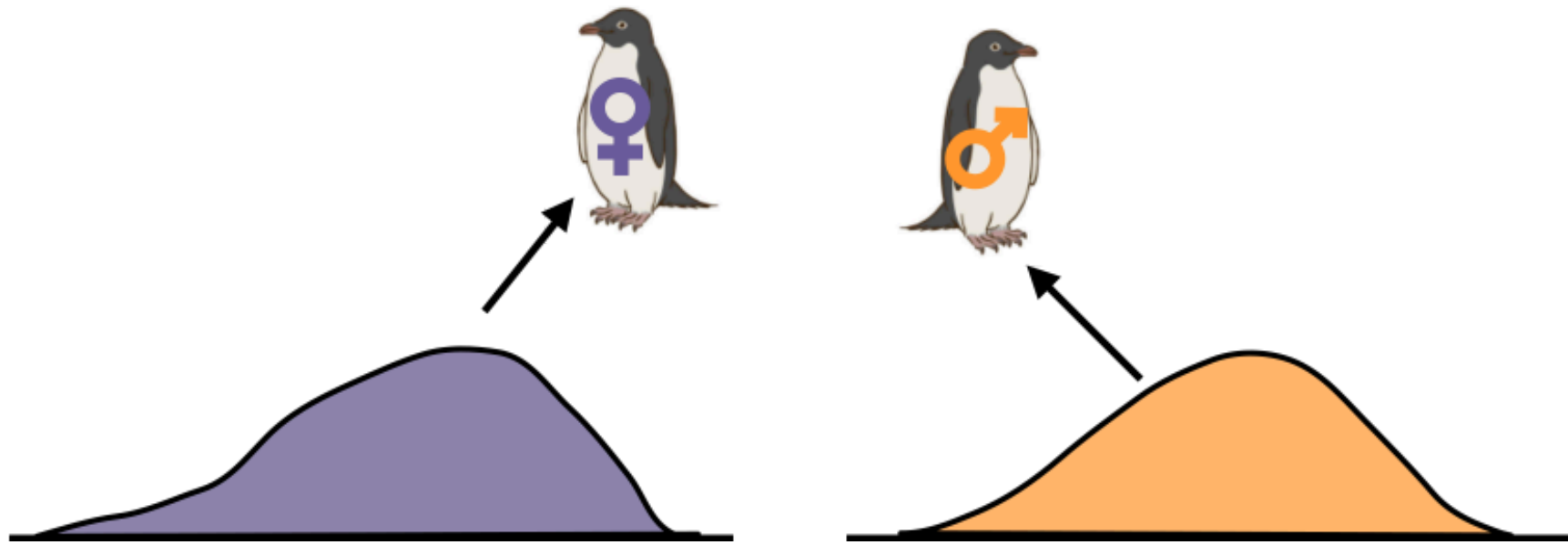
- Feel free to copy most of the code from the previous example. Just make sure to change the data to `gentoo` and the summary to `median()`
- In many of the experimental outcomes, the median was in fact 215. So this result doesn't seem to contradict the claim of the median being 215.

Now, consider comparing *two* populations.

How does the median body mass compare between male and female Adelie penguins?

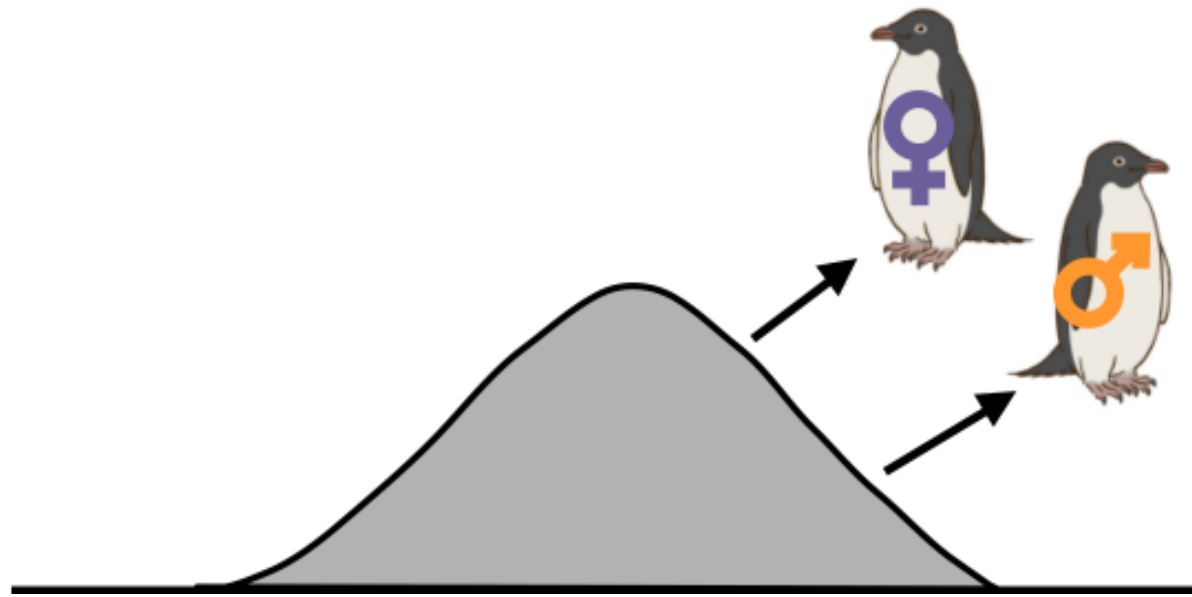
More specifically, is the median the same, or different?

We know how to resample separately for each population.
Assuming the populations are separate:



We can sample 73 separately from each sub-population, and create two separate median sampling distributions.

Alternatively, what if we assume body mass for both sexes is drawn from the *same* population?

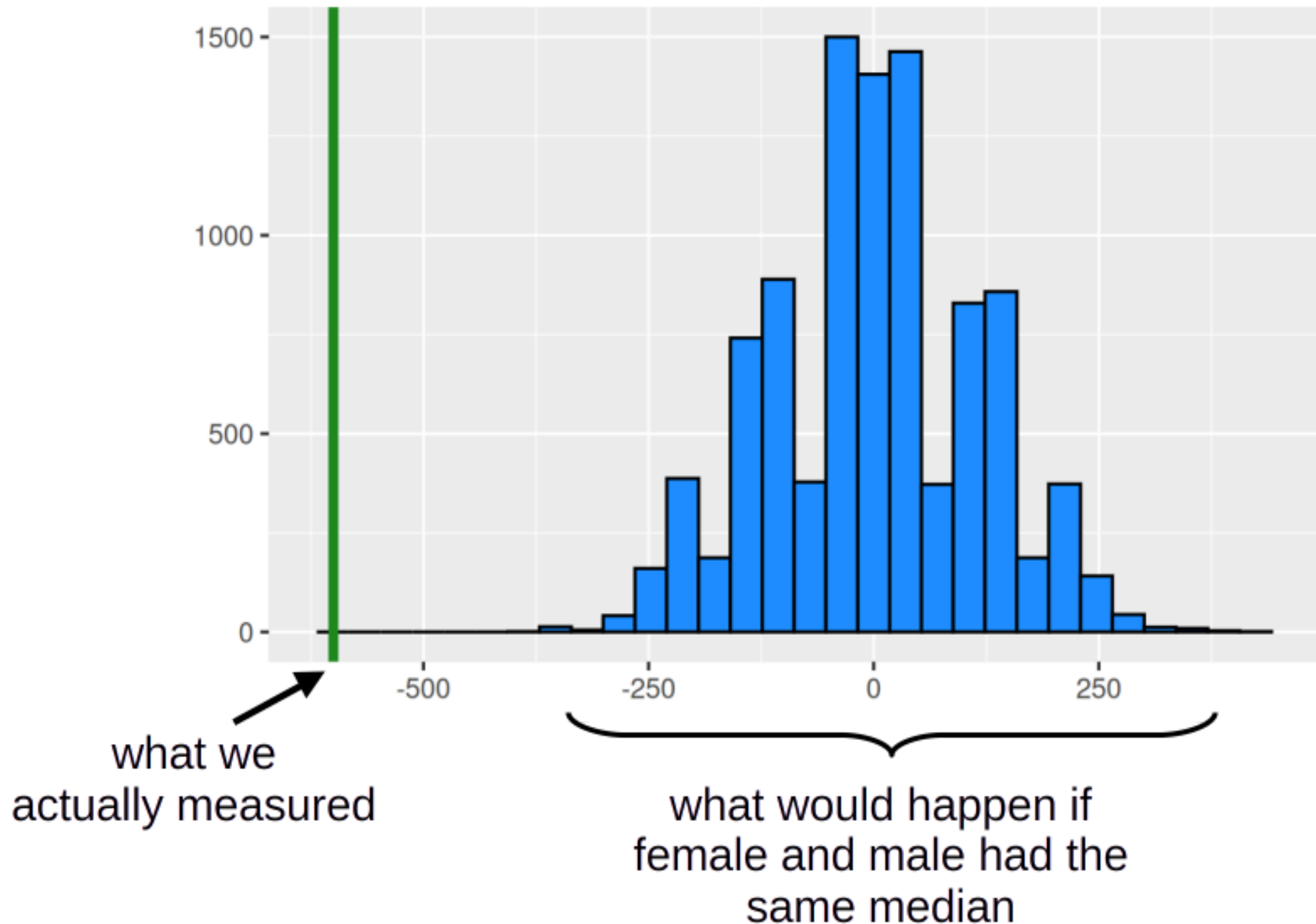


Resampling looks a little bit different.

Start by putting all the data in one big “box”.

Then, randomly resample a new male dataset and a new female dataset, both with replacement from the big box.

Finally, compare the resampled differences in medians to the observed differences in medians.



Exercises

Let's snap back to reality - what does this result actually tell us about physical penguin characteristics by sex?

- Does it look like male and female Adelie penguins have the same median body mass, or different?
- Can we be completely certain of this conclusion?
- CHALLENGE: What would the histogram look like if we resampled the male and female groups separately, *then* calculated the difference in median? How might this other thing be useful?



Solutions/Hints

- Our observed result is way outside the range of the histogram of resamples. It looks like the male and female penguins have different median body mass.
- We can't be completely sure, since this experiment is based on a sample of data, not the full population.
- Looking at a histogram of female and male medians resampled separately would let us learn about the possible *value* of the difference in medians. The histogram would probably be centered around -600.

Looking ahead

In [chapter 7](#), we'll learn the basics of statistical inference with confidence intervals and p-values. Consider the following example:

A factory will shut down for emergency cooling if the average temperature of its machines is over 104.5 degrees F. A foreman measures the machines and calculates an average temperature of 103.7 degrees.

Should the factory be shut down? What are the two ways we could make a “wrong decision”, and which is worse?